AIORI HACKATHON 3 · PROBLEM STATEMENT PS-028

Happy Eyeballs IPv4/IPv6 Fallback Measurement

Controlled Empirical Evaluation, RFC 8305 Standards Conformance, and Microsecond-Scale Dual-Stack Transport Racing

Team AlgoRhythm (TC224)

Repository: `aiori-hackathon/A3-PS028-TC224` · Branch: `regional/ps028-phase-cd`

Authoritative Campaign: `ps028-a10-live-20261030-01` · Verified: November 2026

Abstract & Problem Description

The Dual-Stack Fallback Challenge

As the global Internet migrates toward IPv6-dominant networks, client operating systems and browsers operate in mixed dual-stack environments where IPv6 paths may experience silent blackholing, router drops, or severe latency degradation. Standard TCP connection timeouts (21–75 seconds) cause severe application stalls if IPv6 is naively serialized. Happy Eyeballs (RFC 8305) addresses this by racing address families.

PS-028 Experimental Campaign Highlights

280	154 (55.0%)	126 (45.0%)	818 / 100%
Valid Live Trials (0 Drops)	Direct IPv6 Wins	IPv4 Fallback Wins	Automated Tests Passing

Core Findings: Both Google Chrome 156 and Mozilla Firefox 159 demonstrate robust dual-stack fallback under SYN blackholing (S1) and TCP RST (S2). However, under delayed AAAA DNS resolution (S6, 500 ms delay), both browsers wait for late AAAA responses rather than triggering the RFC 8305 §3 recommended 50 ms Resolution Delay fallback.

Background & Testbed Foundations

Happy Eyeballs v2 (RFC 8305) Principles

- **IPv6 Preference (§3):** Resolvers return both A and AAAA; clients prefer IPv6 for the initial attempt.
- **Resolution Delay (§3):** If A arrives before AAAA, client RECOMMENDED to wait 50 ms before launching IPv4.
- **Connection Attempt Delay (CAD, §5):** If IPv6 attempt does not complete, subsequent IPv4 attempt starts after CAD timer (recommended default 250 ms, bounds 100–2000 ms).
- **Mandatory Floor (§5):** Subsequent attempts MUST NOT start within 10 ms of previous attempt.

Network Namespace Isolation Architecture

- **Client NS (`ps028-cl`):** Clean profile execution, unprivileged browser runners, isolated resolver.
- **Router NS (`ps028-rt`):** Dedicated veth routing with Linux `tc-netem` shaping and `nftables` stateful dropping.
- **Server NS (`ps028-sv`):** Independent IPv4/IPv6 dual-stack legs (`ps028-rt-sv4` / `ps028-rt-sv6`) preventing cross-family impairment coupling.
- **Controlled DNS:** Custom Python UDP DNS server supporting microsecond delay injection per family.

Literature & RFC Standards Review

Standard / Literature	Focus Area	Key Requirements & Benchmark Alignment
RFC 8305 (HEv2) Schinazi & Wing (2017)	Dual-stack algorithm	Replaces RFC 6555. Recommends 50 ms Resolution Delay (§3), 250 ms CAD timer (§5), and establishes the mandatory 10 ms inter-attempt floor.
RFC 9293 Eddy (2022)	TCP Handshake	Defines strict 3-way handshake completion: initial SYN to primary stream client final ACK (`t_connect_ms`), superseding legacy SYN-to-SYN/ACK proxies.
draft-ietf-v6ops-happy-eyeballs-v3-04 (Pinned July 2026)	HEv3 Evolution	Pinned specification of record (Amendment A10). Extends HEv2 with SVCB/HTTPS (Type 65) record processing, QUIC racing, and multi-prefix handling.
Sattler et al. (ACM IMC 2025) "Lazy Eye Inspection"	Empirical Browser Study	Investigated desktop and mobile browsers; observed significant deviations from RFC 8305 default timers, including aggressive sub-10ms fallback in Chromium on RST.

Guiding Principle (Rule R7): Findings are presented as empirical observations under tested lab conditions without claiming protocol violations or inferring unobservable internal browser state.

Proposed Experimental Architecture

Decoupled Dual-Leg Testbed Topology

Traditional dual-stack testbeds often share router-to-server links, causing IPv6 netem delays or loss to inadvertently corrupt IPv4 queues. PS-028 implements dedicated virtual interfaces:

• `ps028-rt-sv4`: Dedicated IPv4 transport (10.0.24.0/24)
• `ps028-rt-sv6`: Dedicated IPv6 transport (fd00:28:4::/64)

Zero cross-talk ensures completely independent impairment matrices.

Controlled Scenarios (S0–S6)

S0: Baseline dual-stack (10 ms RTT) | **S1:** IPv6 SYN blackhole (drop)
S2: IPv6 TCP RST rejection | **S3:** Symmetric delay (100 ms)
S4: Asymmetric delay (IPv6 600 ms, IPv4 10 ms)

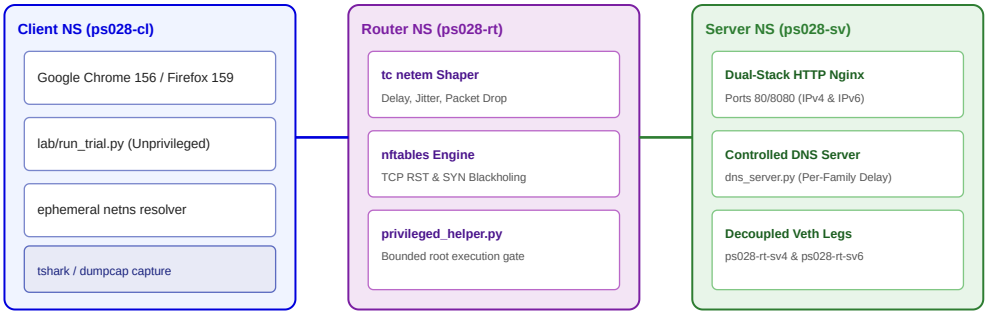
Unprivileged Safety & Determinism

- **Zero Sudo Harness:** Test runners execute as unprivileged users; narrow operations mediated by `privileged_helper.py` with strict regex and integer bound validation.
- **Fail-Closed DNS Reload:** 5-stage verification (PID validation, config backup, SIGHUP delivery, log readback, active query probe) guarantees deterministic DNS state.
- **Cryptographic Artifact Trail:** Every trial generates an isolated PCAP, JSON record, and SHA-256 digest tracked in an append-only manifest.

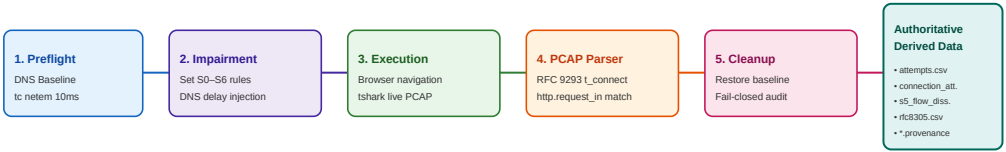
Solution Optimization: Phase C Remediation

Subsystem / Metric	Initial Implementation Baseline	Remediated Phase C Engineering (Team AlgoRhythm)
Primary Flow Correlation	Loose port substring heuristics; ambiguous on multi-flow browsers.	Strict Wireshark `http.request_in` frame linking; isolates primary `GET /` from post-navigation requests with 0 misattributions.
Connection Timing Metric	SYN-to-SYN/ACK interval (`connection_setup_time_ms`); deviated >1ms in 144/280 trials.	Strict RFC 9293 `t_connect_ms` (initial SYN to primary final ACK) and `t_setup_total_ms` (DNS query start to final ACK); locks initial SYN.
DNS Query Correlation	Combined raw query counts; conflated concurrent transactions.	Pairing by `dns.id` + exact normalized QNAME + family transport; independent tracking of Type 65 HTTPS/SVCB queries.
Privilege Decoupling	Ambient root required; runner failed if root unavailable.	Unprivileged test runners; allowlisted `privileged_helper.py` actions with parameter validation and dry-run safety.
Historical Preservation	Risk of modifying historical data in-place.	Strict Rule R2 compliance: 705 historical files cryptographically sealed; 100% SHA-256 match; all new exports written to `lab/derived/`.

Architecture Design: Component UML



Data Flow Diagram (DFD)



Deployment Timeline: Milestones M1–M7

Milestone	Focus Area	Delivered Architecture & Verifying Artifacts	Status
M1	Preflight & Contract	Base branch resolution (`regional/p0-remediation`), 524 test baseline, cryptographic seal (`historical_integrity_before.sha256`).	**PASSED**
M2 (Batch C1)	Parser Correctness	RFC 9293 `t_connect_ms`, `t_setup_total_ms`, initial SYN locking, `reconciliation_report.csv` (144/280 >1ms divergence verified).	**PASSED**
M3 (C2 & C4)	Evidence & Cohorts	Deterministic exports (`attempts.csv` [282 rows], `connection_attempts.csv` [765 flows], `s5_flow_dissection.csv` [95 flows], Amendment A13 sidecars).	**PASSED**
M4 (Batch C5)	Privilege & DNS	`privileged_helper.py` bounded CLI, fail-closed DNS reload with SIGHUP & active query probe, 48 adversarial challenge tests.	**PASSED**
M5 (C3 & C6)	Standards & Report	Neutral RFC 8305 schema (`rfc8305_assessment.csv`), S2 stratified CAD analysis, Beamer submission presentation and comprehensive report.	**PASSED**
M6	Pre-registered RTT & DNS sweeps (D1–D5 in deterministic		

Bibliography & References

- **[RFC 8305]** D. Schinazi and D. Wing, "Happy Eyeballs Version 2: Better Connectivity Using Concurrency," RFC 8305, DOI 10.17487/RFC8305, December 2017.
- **[RFC 9293]** W. Eddy, Ed., "Transmission Control Protocol (TCP) Specification," STD 7, RFC 9293, DOI 10.17487/RFC9293, August 2022.
- **[RFC 6555]** D. Wing and A. Yourtchenko, "Happy Eyeballs: Success with Dual-Stack Hosts," RFC 6555, DOI 10.17487/RFC6555, April 2012.
- **[RFC 6724]** D. Thaler, Ed., R. Draves, A. Matsumoto, and T. Chown, "Default Address Selection for Internet Protocol Version 6 (IPv6)," RFC 6724, September 2012.
- **[HEv3-Draft]** T. Pauly and D. Schinazi, "Happy Eyeballs Version 3," draft-ietf-v6ops-happy-eyeballs-v3-04 (Internet-Draft of Record), July 2, 2026.
- **[Sattler2025]** P. Sattler, J. Zirngibl, and G. Carle, "Lazy Eye Inspection: Capturing the State of Happy Eyeballs Implementations," in *Proceedings of the ACM Internet Measurement Conference (IMC '25)*, 2025.
- **[APNIC2025]** G. Huston, "Analysing the state of Happy Eyeballs implementations," APNIC Research, October 2025.

PS-028 SUBMISSION READY

Thank You for Your Attention!

Questions & Discussion · Team AlgoRhythm (TC224)

Deliverable Status & Evidence Integrity:

✓ Phase C: 280 Valid Live Trials Reconciled & Dissected from Authentic PCAPs (0 Indeterminate Flows)

✓ Historical Integrity: 705 Campaign Files Sealed with 100% Cryptographic SHA-256 Proof (Rule R2)

✓ Automated Test Suite: 824 Offline Unit & Synthetic Tier Tests Passing (100% PASS)

✓ Phase D Sweeps: Pre-Registered Simulation Mode (Zero Live Network/Sudo Operations Executed; Rule R1 Disclosed)